Practical Exercise Worksheet

Disk Imaging with dd

Learning Objectives

• Understand what forensic imaging is and why it is important

• Safely use the dd command to create a forensic image

• Explain the purpose of parameters such as bs, noerror and sync

• Reflect on risks and safeguards when handling storage devices

Important Safety Warning

This exercise uses low-level disk access commands. If you choose the wrong device, you may permanently destroy data on your computer.

Only use a USB stick you own and are willing to erase. If you are unsure, stop and ask for help.

Task 1: Identify Your USB Device

Insert your USB stick and run:

```
lsblk
```

or

```
sudo fdisk -l
```

Identify which device corresponds to your USB drive (e.g. /dev/sdb) and which is your system drive (e.g. /dev/sda).

Your USB device name: _____

Task 2: Create a Forensic Image of the USB

Use the following command structure to image your USB stick safely:

```
sudo dd if=/dev/sdX of=~/usb_image.img bs=128K conv=noerror,sync
```

Replace /dev/sdX with the device name of your USB stick (for example /dev/sdb). This will create a file called usb_image.img in your home directory.

What the Parameters Mean

Parameter	**Explanation**
if=	Input file (the evidence source)
of=	Output file (the forensic image)
bs=128K	Block size used during copying
conv=noerror	Continue imaging even if read errors occur
conv=sync	Pads unreadable blocks with zeroes to preserve data offsets

Task 3: Conceptual Exercise (Do NOT run)

The following command would image the main system disk /dev/sda. You must **not run** this on your own machine.

sudo dd if=/dev/sda of=~/disk_image.img bs=128K conv=noerror,sync

Reflection questions:

- Why is imaging /dev/sda more risky than imaging a USB stick?
- What permissions and safeguards would be required in a real investigation?

Questions for Reflection

1. What is the purpose of forensic imaging?

Your answer:

2. Why are conv=noerror and conv=sync important in forensic contexts?

Your answer:

3. What could happen if you accidentally reverse if= and of=?

Your answer:

4. Why is block size a forensic consideration, not just a technical one?

Your answer:

5. Would you trust a forensic tool that does not document its imaging behaviour? Why or why not?

Your answer: